Mathematical Manuscripts
Overview and Verification Guide

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Abstract

Fourteen research manuscripts in ten result families, distinguishing analytic arguments, Lean kernel-checked supporting lemmas, numerical diagnostics and physical interpretations. External peer review is not implied.

Flagship: fourth-order cubic–quintic variational PDE

On $H^2(\mathbb{R}^2;\mathbb{C})$, for $k,\beta,\gamma>0$, consider

$$H(u)=\|(\Delta+k^2)u\|_2^2-\frac{\beta}{2}\|u\|_4^4+\frac{\gamma}{3}\|u\|_6^6.$$

The primary family presents sufficient conditions for strict subadditivity, minimizer existence and orbital stability in the complete minimizer set. These claims concern a declared independent mathematical Hamiltonian.

Result families

Family	Contents
F01	Strict binding, minimizers, orbital stability and negative trials
F02	Higher-derivative constraint counting and Sobolev UV obstruction
F03	Spectral–phase exact identity and closed-curve counterexample
F04	Nonlinear finite Fourier subspace non-invariance
F05	Positive curvature whole-space nonattainment
F06	Finite-wavenumber dispersion minimum
F07	Exact weighted phase-locking variational result
F08	Rank and signature constraints on scalar-gradient tensors
F09	11-by-31 structural local identifiability obstruction
F10	No L^2 state supported on a single Fourier shell

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Browse the catalogue and PDFs at https://github.com/rickyjreyes/math/blob/main/CONTENTS.md. Lean declarations and explicit verification status: https://github.com/rickyjreyes/math/blob/main/lean/formalization.yaml. The included Lean code does not formalize the main nonlinear-PDE theorems.